

HES 2308 MEDICAL DATA ANALYSIS II

Lecture Notes
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Introduction

Survival analysis is the phrase used to describe the analysis of data in the form of times from a well defined time origin until the occurrence of some particular event or end point.

If the end point is the death of a patient the resulting data are literally survival times.

Special Features of Survival Data

The main features are:

- i. Survival data is positively skewed
- ii. They are always positive
- iii. Presence of censoring

Censoring

The survival time of an individual is said to be censored when the end point of interest has not been observed for that individual.

This is caused by:

- i. Data from the study are to be analysed at a point in time when some individuals are still alive
- ii. The survival status of the individual may be lost to follow up by the time of the study.
- iii. Death from a cause that is unrelated to the treatment.

A patient who entered a study at time t_0 dies at time $t_0 + t$.

However time t is unknown either because the individual is still alive or because he or she has been lost to follow up.

There are different types of censoring in survival analysis.

A patient who entered a study at time t_0 dies at time $t_0 + t$.

However, t is unknown either because the individual is still alive or because he or she has been lost to follow up.

If the individual was last known to be alive at time $t_0 + c$ the time c is called a censored survival time.

Right Censoring This censoring occurs after an individual has been entered into a study that is to the right of the last known survival time and is therefore known as *right censoring*.

The right-censored survival time is then less than the actual but unknown survival time.

Left Censoring This type of censoring is encountered when the actual survival time of an individual is less than that observed.

Interval Censoring Here individuals are known to have experienced an event within an interval of time.

Independent Censoring An important assumption that will be made in the analysis of censored survival data is that the actual survival time of an individual t does not depend on any mechanism that causes that individual's survival time to be censored at time c where $c < t$.

Such censoring is called independent or non-informative censoring.

Survivor, Hazard and Cumulative Hazard Functions

The survivor function

The actual survival time of an individual t can be regarded as the observed value of a random variable T that can take any non-negative value.

The different values of T can take have a *probability distribution* and we call T the *random variable* associated with the survival time.

Now suppose that this random variable has a probability distribution with underlying *probability density function* $f(t)$. The *distribution function* of T is given by:

$$F(t) = P(T < t) = \int_0^t f(u) du \quad (1)$$

and represents the probability that the survival time is less than some value t .

This function is also called the *cumulative incidence function* since it summarises the cumulative probability of death occurring before time t

The survivor function, $S(t)$ is defined to be the probability that the survival time is greater than or equal to t and so from Equation 1;

$$S(t) = P(T \geq t) = 1 - F(t) \quad (2)$$

The survivor function can therefore be used to represent the probability that an individual survives beyond any given time.

The Hazard Function

The hazard function is widely used to express the risk or hazard of an event such as death occurring at some time t .

This function is obtained from the probability that an individual dies at time t conditional on he or she having survived to that time.

Consider the probability that the random variable associated with an individual's survival time T lies between t and $t + \delta t$ conditional on T being greater than or equal to t written $P(t \leq T < t + \delta t | T \geq t)$.

This conditional probability is then expressed as a probability per unit time by dividing by the time interval δt to give a rate.

The hazard function $h(t)$ is then the limiting value of this quantity as δt tends to zero, so that

$$h(t) = \lim_{\delta t \rightarrow 0} \left\{ \frac{P(t \leq T < t + \delta t | T \geq t)}{\delta t} \right\} \quad (3)$$

The function $h(t)$ is also referred to as *hazard rate*, the *instantaneous death rate*, *intensity rate*, *force of mortality*.

The conditional probability in the definition of the hazard function in equation 3 is given by

$$\frac{P(t \leq T < t + \delta t)}{P(T \geq t)}$$

which is equal to

$$\frac{F(t + \delta t) - F(t)}{S(t)}$$

where $F(t)$ is the distribution function of T . Then

$$h(t) = \lim_{\delta t \rightarrow 0} \left\{ \frac{F(t + \delta t) - F(t)}{\delta t} \right\} \frac{1}{S(t)}$$

Now

$$\lim_{\delta t \rightarrow 0} \left\{ \frac{F(t + \delta t) - F(t)}{\delta t} \right\}$$

is the definition of the derivative of $F(t)$ with respect to t which is $f(t)$ and so

$$h(t) = \frac{f(t)}{S(t)} \quad (4)$$

The cumulative hazard function

From

$$h(t) = \frac{f(t)}{S(t)}$$

it follows that:

$$h(t) = -\frac{d}{dt} \{\log S(t)\} \quad (5)$$

and so

$$S(t) = \exp\{-H(t)\} \quad (6)$$

where

$$H(t) = \int_0^t h(u) du \quad (7)$$

The cumulative hazard function can also be obtained from the survivor function since

$$H(t) = -\log S(t) \quad (8)$$

The cumulative hazard function $H(t)$ is the cumulative risk of an event occurring by time t .

Parametric Survival Distributions

Several survival distributions exist for modeling survival data.

Exponential Distribution

The p.d.f is given as:

$$f(t) = \lambda e^{-\lambda t}$$

The hazard function

$$h(t) = \lambda$$

The cumulative hazard function is given as:

$$H(t) = \int_0^t h(u) du = \int_0^t \lambda du = \lambda t \Big|_0^t = \lambda t$$

The survival function:

$$S(t) = \exp^{-H(t)} = e^{-\lambda t}$$

The mean of an exponential distribution is given by:

$$E(T) = \int_0^\infty S(t) dt = \int_0^\infty e^{-\lambda t} dt = \frac{1}{\lambda}$$

The median is the value of t that satisfies

$$0.5 = e^{-\lambda t}$$

so that

$$t_{med} = \frac{\log 2}{\lambda}$$

Weibull Distribution

The constant hazard of the exponential distribution makes it difficult to work with as it is not appropriate for describing lifetimes of humans or animals.

The weibull distribution which has a decreasing and increasing hazard rate is more appropriate.

The hazard function is given as:

$$h(t) = \alpha \lambda^\alpha t^{\alpha-1} \quad (9)$$

The cumulative hazard and survival function are given respectively by:

$$H(t) = (\lambda t)^\alpha$$

and

$$S(t) = e^{-(\lambda t)^\alpha}$$

The mean and median of the weibull distribution are given respectively by:

$$E(T) = \frac{\gamma(1 + 1/\alpha)}{\lambda}$$

and

$$t_{med} = \frac{[\log(2)]^{1/\alpha}}{\lambda}$$

Some Non-Parametric Procedures

Introduction

These methods are said to be non-parametric or distribution free since they do not require specific assumptions to be made about the underlying distribution of the survival times.

Suppose first that we have a single sample of survival times, where none of the observations are censored.

The survivor function $S(t)$ is the probability that an individual survives for a time greater than or equal to t .

The function can be estimated by the *empirical survivor function* given by:

$$\hat{S}(t) = \frac{\text{Number of individuals with survival times} \geq t}{\text{Number of individuals in the data set}} \quad (10)$$

Equivalently, $\hat{S}(t) = 1 - \hat{F}(t)$ where $\hat{F}(t)$ is the *empirical distribution function* that is the ratio of the total number of individuals alive at time t to the total number of individuals in the study.

The estimated survivor function $\hat{S}(t)$ is assumed to be constant between two adjacent death times and so a plot of $\hat{S}(t)$ against t is a step function.

The function decreases immediately after each observed survival time.

Example

The following are survival times, in months of eleven male patients:

11, 13, 13, 13, 13, 14, 14, 15, 15, 17

Compute the survival time.

Solution

The estimated values of the survivor function at times 11, 13, 14, 15 and 17 months are 1.000, 0.909, 0.455, 0.273 and 0.091.

This method cannot be used when there are censored observations.

Life-Table Estimate of the Survivor Function

The life table estimate of the survivor function also known as the actuarial estimate of the survivor function is obtained by first dividing the period of observations into a series of time intervals.

Suppose that the j - th of m such intervals, $j = 1, 2, \dots, m$ extends from time t'_{j-1} to immediately before time t'_j where we take $t_0 = 0$ and $t_m = \infty$.

Also let d_j and c_j denote the number of deaths and the number of censored survival times, respectively in this interval and let n_j be the number of individuals who are alive and therefore at risk of death at the start of the j - th interval.

We now make the assumption that the censoring process is such that the censored survival times occur uniformly throughout the j - th interval so that the average number of individuals who are at risk during this interval is;

$$n'_j = n_j - c_j/2 \quad (11)$$

In the j th interval the probability of death can be estimated by $\frac{d_j}{n'_j}$ so that the corresponding survival probability is $\frac{n'_j - d_j}{n'_j}$

The life-table estimate of the survival function is given by:

$$S^*(t) = \prod_{i=1}^{j-1} \left(\frac{n'_i - d_i}{n'_i} \right) \quad (12)$$

for $t'_{j-1} \leq t < t'_j, j = 2, 3, \dots, m$

Interval	Time Period	d_j	c_j	n_j	n'_j	$(n'_j - d_j)/n'_j$	$S^*(t)$
1	0	16	4	48	46	0.6522	1.000
2	12	10	4	28	26	0.6154	0.6522
3	24	1	0	14	14	0.9286	0.4013
4	36	3	1	13	12.5	0.7600	0.3727
5	48	2	2	9	8.0	0.7500	0.2832
6	60	4	1	5	4.5	0.1111	0.2124

Kaplan-Meier Estimate of the Survival Function

Suppose that there are n individuals with observed survival times $t_1, t_2, \dots, t_n$. Some of these observations may be right-censored and there may also be more than one individual with the same observed survival time.

We therefore suppose that there are r death times amongst the individuals where $r \leq n$.

After arranging these death times in ascending order, the j th is denoted $t_{(j)}$ for $j = 1, 2, \dots, r$ and so the r ordered death times are $t_{(1)} < t_{(2)} < \dots < t_{(r)}$

The number of individuals who are alive just before time $t_{(j)}$ including those who are about to die at this time will be denoted by n_j for $j = 1, 2, \dots, r$ and d_j will denote the number of individuals who die at this time.

The time interval from $t_{(j)} - \delta$ to $t_{(j)}$ where δ is an infinitesimal time interval, then includes one death time.

Since there are n_j individuals who are alive just before $t_{(j)}$ and d_j deaths at $t_{(j)}$ the probability that an individual dies during the interval from $t_{(j)} - \delta$ to $t_{(j)}$ is estimated by $\frac{d_j}{n_j}$.

The corresponding estimated probability of survival through that interval is then $\frac{n_j - d_j}{n_j}$

The Kaplan-Meier estimate of the survivor function is given by:

$$\hat{S}(t) = \prod_{j=1}^k \left(\frac{n_j - d_j}{n_j} \right) \quad (13)$$

for $t_{(k)} \leq t < t_{(k+1)}$, $k = 1, 2, \dots, r$ with $\hat{S}(t) = 1$ for $t < t_{(1)}$ and where $t_{(r+1)}$ is taken to be ∞

Example 1

We will start with the `aml` dataset in the `survival` package in R. The data gives time to death for patients with acute myelogenous leukemia, comparing "maintained" chemotherapy treatment to "nonmaintained". The variables are:

- `time`: survival or censoring time
- `status`: 0=censored, 1= death
- `x` chemotherapy maintained or non-maintained

```
library(survival)
# ~ 1 indicates KM survival function estimate for whole sample
```


time	status
9	1
13	1
13	0
18	1
23	1
28	0
31	1
34	1
45	0
48	1
161	0
5	1
5	1
8	1
8	1
12	1
16	0
23	1
27	1
30	1
33	1
43	1
45	1

```
KM <- survfit(Surv(time, status) ~ 1, data=aml)
# save KM survival function as tibble (modern data.frame)
KM.tab <- broom::tidy(KM)
KM.tab # same as print(KM.tab)
```

A tibble: 18 x 8

	time	n.risk	n.event	n.censor	estimate	std.error	conf.high	conf.low
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	5	23	2	0	0.913	0.0643	1	0.805
2	8	21	2	0	0.826	0.0957	0.996	0.685
3	9	19	1	0	0.783	0.110	0.971	0.631
4	12	18	1	0	0.739	0.124	0.942	0.580
5	13	17	1	1	0.696	0.138	0.912	0.531
6	16	15	0	1	0.696	0.138	0.912	0.531
7	18	14	1	0	0.646	0.157	0.878	0.475
8	23	13	2	0	0.547	0.196	0.803	0.372
9	27	11	1	0	0.497	0.218	0.762	0.324
10	28	10	0	1	0.497	0.218	0.762	0.324
11	30	9	1	0	0.442	0.248	0.718	0.272
12	31	8	1	0	0.386	0.282	0.671	0.223
13	33	7	1	0	0.331	0.321	0.622	0.177
14	34	6	1	0	0.276	0.369	0.569	0.134
15	43	5	1	0	0.221	0.432	0.515	0.0947
16	45	4	1	1	0.166	0.519	0.458	0.0598
17	48	2	1	0	0.0828	0.877	0.462	0.0148
18	161	1	0	1	0.0828	0.877	0.462	0.0148

Example 2

Calculate the Kaplan Meier Estimate for the following data:

n_j	d_j	$(n_j - d_j)/n_j$	$\hat{S}(t)$
18	0	1.0000	1.0000
18	1	0.9444	0.9444
15	1	0.9333	0.8815
13	1	0.9231	0.8137
12	1	0.9167	0.7489
8	1	0.8750	0.6526
7	1	0.8571	0.5594
6	1	0.8333	0.4662
5	1	0.8000	0.3729
3	1	0.6667	0.2486

Standard Error of the Kaplan-Meier Estimate

The Kaplan-Meier estimate of the survivor function for any value of t in the interval from $t_{(k)}$ to $t_{(k+1)}$ can be written as:

$$\hat{S}(t) = \prod_{j=1}^k \hat{p}_j$$

for $k = 1, 2, \dots, r$ where $\hat{p}_j = \frac{n_j - d_j}{n_j}$ is the estimated probability that an individual survives through the time interval that begins at $t_{(j)}$, $j = 1, 2, \dots, r$. Taking logarithms

$$\log \hat{S}(t) = \sum_{j=1}^k \log \hat{p}_j$$

and so the variance of $\log \hat{S}(t)$ is given by:

$$\text{var}\{\log \hat{S}(t)\} = \sum_{j=1}^k \text{var}\{\log \hat{p}_j\} \quad (14)$$

Since $\hat{p}_j = (n_j - d_j)/n_j$ the variance of $\hat{p}_j$ is $\text{var}(n_j - d_j)/n_j^2$ that is $p_j(1 - p_j)/n_j$

The variance of $\hat{p}_j$ can be estimated by:

$$\hat{p}_j(1 - \hat{p}_j)/n_j \quad (15)$$

In order to obtain the variance of $\log \hat{p}_j$ we make use of a general result for the approximate variance of a function of a random variable.

According to this result, the variance of a function $g(X)$ of the random variable X is given by:

$$\text{var}\{g(X)\} \approx \left\{ \frac{dg(X)}{dX} \right\}^2 \text{Var}(X) \quad (16)$$

This is known the Taylor series approximation to the variance of a function of a random variable.

Using the above equation the approximate variance of $\log \hat{p}_j$ is $(1 - \hat{p}_j)/(n_j \hat{p}_j)$ which on substitution for $\hat{p}_j$ reduces to:

$$\frac{d_j}{n_j(n_j - d_j)} \quad (17)$$

Then

$$\text{var}\{\log \hat{S}(t)\} \approx \sum_{j=1}^k \frac{d_j}{n_j(n_j - d_j)} \quad (18)$$

and a further application of the result gives:

$$\text{var}\{\log \hat{S}(t)\} \approx \frac{1}{[\hat{S}(t)]^2} \text{var}\{\hat{S}(t)\}$$

so that:

$$\text{var}\{\hat{S}(t)\} \approx [\hat{S}(t)]^2 \sum_{j=1}^k \frac{d_j}{n_j(n_j - d_j)} \quad (19)$$

Finally the standard error of the Kaplan-Meier estimate of the survivor function defined by the square root of the estimated variance of the estimate is given by:

$$\text{se}\{\hat{S}(t)\} \approx \hat{S}(t) \left\{ \sum_{j=1}^k \frac{d_j}{n_j(n_j - d_j)} \right\}^{\frac{1}{2}} \quad (20)$$

for $t_{(k)} \leq t < t_{(k+1)}$. This result is known as *Greenwood's formula*

Nelson-Aalen Estimate of the Survivor Function

An alternative estimate of the survivor function which is based on the individual event time is the Nelson-Aalen estimate given by:

$$\bar{S}(t) = \prod_{j=1}^k \exp\left(-\frac{d_j}{n_j}\right) \quad (21)$$

We use the values in Example 2 to calculate the Nelson-Aalen Estimate

$\exp(-d_j/n_j)$	$\bar{S}(t)$
1.0000	1.0000
0.9460	0.9460
0.9355	0.8850
0.9260	0.8194
0.9200	0.7539
0.8825	0.6653
0.8669	0.5768
0.8465	0.4882
0.8187	0.3997
0.7165	0.2864

The standard error of the Nelson-Aalen estimate is

$$se\{\hat{S}(t)\} \approx \hat{S}(t) \left\{ \sum_{j=1}^k \frac{d_j}{n_j^2} \right\}^{\frac{1}{2}}$$

Maximum Likelihood Estimation

If we know that a random variable T has an exponential distribution with parameter $\lambda = 0.03$ we can directly compute the probability that T exceeds a particular value.

Suppose that we have a series of observations $t_1, t_2, \dots, t_n$ from an exponential distribution with unknown parameter λ , then we can estimate the parameter λ via maximum likelihood estimation.

Assuming that there is no censoring, the likelihood takes the general form

$$L(\lambda; t_1, t_2, \dots, t_n) = f(t_1, \lambda) \times f(t_1, \lambda) \times \dots \times f(t_n, \lambda) = \prod_{i=1}^n f(t_i, \lambda) \quad (22)$$

If some observations are censored, we have to make adjustments.

For an observation of an observed death, we put in the pdf.

But for a right-censored observation we put in the survivor function indicating that the observations is known only to exceed a particular value.

The likelihood in general takes the form:

$$L(\lambda; t_1, \dots, t_n) = \prod_{i=1}^n f(t_i, \lambda)^{\delta_i} S(t_i, \lambda)^{(1-\delta_i)} = \prod_{i=1}^n h(t_i, \lambda)^{\delta_i} S(t_i, \lambda)$$

The expression means that when t_i is an observed death, the censoring indicator is $\delta_i = 1$ and we enter the pdf factor.

When t_i is a censored observation we have $\delta_i = 0$ and we enter the survival function.

For the case of exponential distribution we substitute the expressions for the pdf and survival functions and simplify as follows:

$$L(\lambda) = \prod_{i=1}^n [\lambda e^{-\lambda t_i}]^{\delta_i} [e^{-\lambda t_i}]^{1-\delta_i} = \lambda^d e^{-\lambda V} \quad (23)$$

We have the total number of deaths

$$d = \sum_{i=1}^n \delta_i$$

and the total amount of time of patients on the study

$$V = \sum_{i=1}^n t_i$$

We can maximize the log-likelihood as follows:

$$l(\lambda) = d \log \lambda - \lambda V$$

Getting first derivative and equating to zero we get

$$\hat{\lambda} = \frac{d}{V}$$

The second derivative of the log-likelihood

$$l'' = -\frac{d}{\lambda^2} = -I(\lambda)$$

Using Standard mathematical statistics theory, the inverse of the information matrix is approximately the variance of the m.l.e

$$Var(\hat{\lambda}) \approx I^{-1}(\lambda) = \frac{\lambda^2}{d} = \frac{d}{V^2}$$

Comparison of Two Groups of Survival Data

The simplest way of comparing the survival times obtained from two groups of individuals is to plot the corresponding estimates of the two survivor functions on the same axes.

In the comparison of two groups of survival times, there are number of methods that can be used to quantify the extent of between group differences.

Log-Rank Test

In order to construct the log-rank test we begin by considering separately each death time in two groups of survival data.

These groups will be labelled Group I and II. Suppose that there are r distinct death times, denoted by $t_{(1)} < t_{(2)} < \dots < t_{(r)}$ across the two groups and that at time $t_{(j)}$, d_{1j} individuals and d_{2j} individuals in Group II die, for $j = 1, 2, \dots, r$.

Suppose further that there are n_{1j} individuals at risk of death in the first group just before time $t_{(j)}$ and that there are n_{2j} individuals at risk in the second group.

Consequently at time $t_{(j)}$ there are $d_j = d_{1j} + d_{2j}$ deaths in total out of $n_j = n_{1j} + n_{2j}$ individuals at risk.

The situation is summarized below:

Group	Number of deaths at $t_{(j)}$	Number Surviving Beyond $t_{(j)}$	Number at Risk Just before $t_{(j)}$
I	d_{1j}	$n_{1j} - d_{1j}$	n_{1j}
II	d_{2j}	$n_{2j} - d_{2j}$	n_{2j}
Total	d_j	$n_j - d_j$	n_j

Consider the null hypothesis that there is no difference in the survival experience of the individuals in the two groups.

d_{1j} is a random variable which can take any value in the range from 0 to the minimum d_j and n_{1j} .

In fact d_{1j} has a distribution known as the *hypergeometric distribution* according to which the probability that the random variable associated with the number of deaths in the first group takes the value d_{1j} is:

$$\frac{\binom{d_j}{d_{1j}} \binom{n_j - d_j}{n_{1j} - d_{1j}}}{\binom{n_j}{n_{1j}}}$$

The mean of the hypergeometric random variable d_{1j} is given by:

$$e_{1j} = \frac{n_{1j} d_j}{n_j}$$

so that e_{1j} is the expected number of individuals who die at time $t_{(j)}$ in Group I.

Summing the differences $d_{1j} - e_{1j}$ over the total number of death times, r in the two groups. The resulting statistic is given by:

$$U_L = \sum_{j=1}^r (d_{1j} - e_{1j}) \quad (24)$$

Since d_{1j} has a hypergeometric distribution the variance of d_{1j} is given by:

$$v_{1j} = \frac{n_{1j}n_{2j}d_j(n_j - d_j)}{n_j^2(n_j - 1)}$$

so that the variance of U_L is;

$$Var(U_L) = \sum_{j=1}^r v_{1j} = V_L$$

Furthermore it can be shown that U_L has an approximate normal distribution when the number of deaths is not too small.

It then follows that $U_L / \sqrt{V_L}$ has a normal distribution with mean zero and unit variance.

We therefore write:

$$\frac{U_L}{\sqrt{V_L}} \sim N(0, 1)$$

The square of a standard normal random variable has a chi-squared distribution on one degree of freedom denoted by χ_1^2 so we have that:

$$\frac{U_L^2}{V_L} \sim \chi_1^2$$

Example

Two groups of patients suffering from malaria fever were asked to report daily at a local hospital, until they had made a complete recovery. One group of six patients was treated with Coast Special medicine, and the other group of six patients was treated with Kisumu Kalmer medicine. The observed number of days to complete recovery was observed for both groups, as shown below, with censored times indicated by +. Treated with Coast Special medicine 1, 3, 5, 7, 11+, 12 Treated with Kisumu Kalmer medicine 1, 8, 10, 13, 14+, 15+

Use the log rank test at the 5% level of significance to determine whether there is a significant difference in the effects of the two treatments.

Solution

The hypothesis of the study:

H_0 : The survival times for the two groups are the same.

H_1 : The survival times for the two groups are not the same.

Death time	d1j	n1j	d2j	n2j	dj	nj	e1j	v1j
1	1	6	1	6	2	12	1	0.454545
3	1	5	0	5	1	10	0.5	0.25
5	1	4	0	5	1	9	0.444444	0.246914
7	1	3	0	5	1	8	0.375	0.234375
8	0	2	1	5	1	7	0.285714	0.204082
10	0	2	0	4	0	6	0	0
11	0	2	0	4	0	6	0	0
12	1	2	0	4	1	6	0.333333	0.222222
13	0	1	1	4	1	5	0.2	0.16
14	0	1	0	3	0	4	0	0
15	0	1	0	3	0	4	0	0
Total		5					3.138492	1.772138

$$U_L = 5 - 3.138 = 1.8615$$

$$V_L = 1.772$$

$$W_L = \frac{1.8615 \times 1.8615}{1.772} = 1.955$$

In R:

```
# reading in the data
library(survival)
time <- c(1,3,5,7,11,12,1,8,10,13,14,15)
status <- c(1,1,1,1,0,1,1,1,1,1,0,0)
group <- c(1,1,1,1,1,1,2,2,2,2,2,2)
#perform log rank test
survdif(Surv(time, status) ~ group, data=ovarian)
```

Call:

```
survdif(formula = Surv(time, status) ~ group, data = ovarian)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
group=1	6	5	3.19	1.029	1.82
group=2	6	4	5.81	0.565	1.82

Chisq= 1.8 on 1 degrees of freedom, p= 0.2

Plotting the curves

```
time <- c(1,3,5,7,11,12,1,8,10,13,14,15)
status <- c(1,1,1,1,0,1,1,1,1,1,0,0)
group <- c(1,1,1,1,1,1,2,2,2,2,2,2)

data <- data.frame(
  time = time,
  status = status,
  group = factor(group)
)
fit <- survfit(Surv(time, status) ~ group, data = data)
```

```
survdif(Surv(time, status) ~ group, data = data)
```

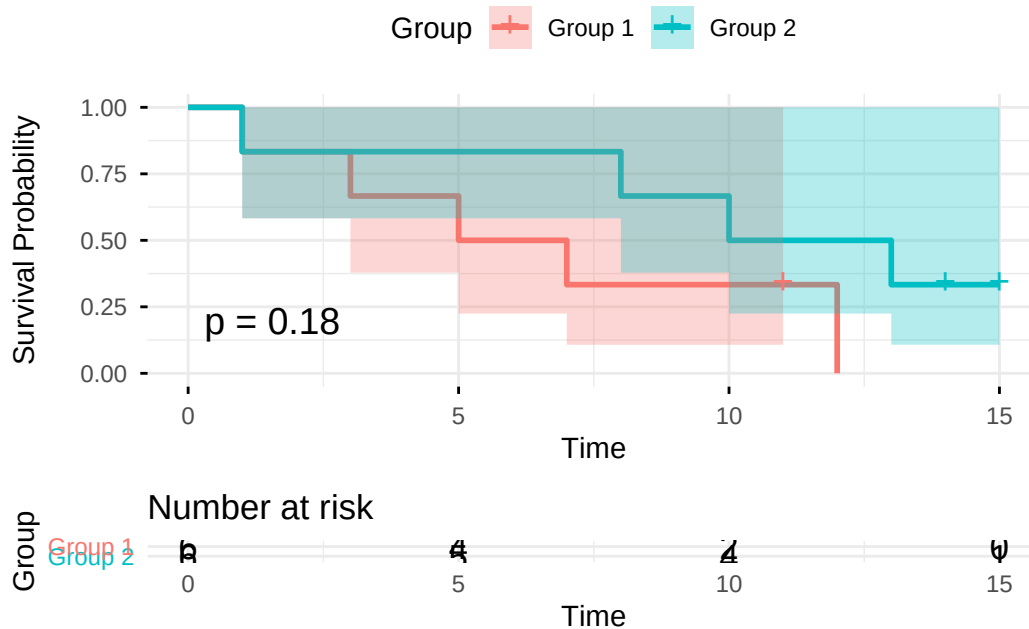
Call:

```
survdif(formula = Surv(time, status) ~ group, data = data)
```

	N	Observed	Expected	$(O-E)^2/E$	$(O-E)^2/V$
group=1	6	5	3.19	1.029	1.82
group=2	6	4	5.81	0.565	1.82

Chisq= 1.8 on 1 degrees of freedom, p= 0.2

```
library(survminer)
ggsurvplot(
  fit,
  data = data,
  conf.int = TRUE,
  pval = TRUE,
  risk.table = TRUE,
  legend.title = "Group",
  legend.labs = c("Group 1", "Group 2"),
  xlab = "Time",
  ylab = "Survival Probability",
  ggtheme = theme_minimal()
)
```



Cox-PH Model

Consider n patients in a study and let $N_i(t) = 1, \dots, n$ be the counting process which counts the event occurrences in the time interval $[0, t]$.

The counting process N_i is associated with the lifetime $T_i, T_i \sim F = 1 - S$ and censoring variable $C_i, C_i \sim G$, and we have $T_i = T_i \wedge C_i, \delta_i = I(T_i = T_i), \forall i = 1, \dots, n$. Let $h(t)$ be the hazard rate associated with S . With $Y_i(t)$ denoting the at-risk process associated with the i -th observation.

We denote the i -th covariate vector by

$$\mathbf{x}_i(t) = (x_{i1}(t), x_{i2}(t), \dots, x_{ip}(t)).$$

With covariates, the hazard rate of the i -th observation is expressed as $h_i(t | \mathbf{x}_i(t))$. The intensity process $h_i(t)$ associated with the counting process $N_i, i = 1, \dots, n$, can be expressed as

$$h_i(t) = Y_i(t)h_i(t | \mathbf{x}_i(t)). \quad (25)$$

For an individual with covariate vector $x_i(t)$ we consider the following form of relationship between the covariate and the hazard rate:

$$h_i(t|x_i) = h_0(t)r(\beta, x_i(t))$$

Here, the baseline $h_0(t)$ is left unspecified and the relative risk function $r(\beta, x_i(t))$ needs further specification.

The choice of $r(\beta, x_i(t)) = \exp\{\beta'x_i(t)\}$ leads to *Cox Proportional hazard* model and is given by:

$$h_i(t|x_i(t)) = h_0(t) \exp\{\beta'x_i(t)\} \quad (26)$$

Cox PH model is also called as exponential relative risk model. Why proportional hazards? Consider two observations $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ where all the components are same except the l -th component and we have

$$x_{2l}(t) = x_{1l}(t) + 1.$$

Then the hazards ratio based on the model 5.7 becomes

$$\frac{h_2(t | \mathbf{x}_2(t))}{h_1(t | \mathbf{x}_1(t))} = \exp\{\beta'(\mathbf{x}_2(t) - \mathbf{x}_1(t))\} = \exp\{\beta_l\}.$$

Clearly, the hazards ratio is now independent of time and given the hazard rate of observation 1 at time t by $h_1(t | \mathbf{x}_1(t))$, we can obtain the second observation $h_2(t | \mathbf{x}_2(t))$ by multiplying the first by a scale of e^{β_l} . Thus, the model is aptly called as the *proportion hazards model*. The intensity process associated with the Cox PH model is thus given by

$$h_i(t) = Y_i(t)h_0(t) \exp\{\beta'\mathbf{x}_i(t)\}, \quad (27)$$

and for the general relative risk model by

$$h_i(t) = Y_i(t)h_0(t)r(\beta, \mathbf{x}_i(t)). \quad (28)$$

Example

Given the following proportional hazards regression model for mortality of a sample of life assurance policy holders.

$$h_i(t) = h_0(t) \exp\{0.01(x_i - 25) + y_i - 0.06z_i\}$$

where

$h_i(t)$ denote the hazard function for life i at duration t

$h_0(t)$ denotes the baseline hazard function at duration t

x_i denotes the age of entry of life i

$y_i = 1$ if life i is a smoker otherwise zero

$z_i = 1$ if life i is a female zero if male.

- Determine the class of the lives to which the baseline hazard function applies
- What does the model tell you about the relative risk of a male smoker aged 25 at entry compared to a male non-smoker aged 40 at entry?
- What does the model tell you about the relative risk of a female non-smoker aged 30 at entry compared to a male smoker aged 35 at entry?

Example in R

Analysis of PBC data under the Cox-PH model. We are interested to see how the variables age, treatment, gender, edema, bilirubin and albumin influence the hazard rate for the patients

```
library(survival)

pbc2<-subset(pbc,select=c(time,status,trt,age,sex, edema,bili,albumin))

pbc2<-na.omit(pbc2)

pbc_ph<-coxph(Surv(time,status==2)~trt+age+sex+ edema+bili+albumin,data=pbc2)

summary(pbc_ph)
```

Call:

```
coxph(formula = Surv(time, status == 2) ~ trt + age + sex + edema +
      bili + albumin, data = pbc2)
```

n= 312, number of events= 125

	coef	exp(coef)	se(coef)	z	Pr(> z)	
trt	-0.014276	0.985825	0.185650	-0.077	0.938705	
age	0.031620	1.032125	0.009406	3.362	0.000775	***
sexf	-0.578187	0.560914	0.247723	-2.334	0.019595	*
edema	1.022151	2.779165	0.316962	3.225	0.001260	**
bili	0.125494	1.133708	0.014851	8.450	< 2e-16	***
albumin	-1.227334	0.293073	0.239965	-5.115	3.14e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
trt	0.9858	1.0144	0.6851	1.4185
age	1.0321	0.9689	1.0133	1.0513
sexf	0.5609	1.7828	0.3452	0.9115
edema	2.7792	0.3598	1.4932	5.1727
bili	1.1337	0.8821	1.1012	1.1672
albumin	0.2931	3.4121	0.1831	0.4691

Concordance= 0.823 (se = 0.02)

Likelihood ratio test= 157.7 on 6 df, p=<2e-16

Wald test = 183.5 on 6 df, p=<2e-16

Score (logrank) test = 280.1 on 6 df, p=<2e-16

Residual Analysis for the Cox-PH Regression Model

Regression analysis plays a paramount role in diagnostics in regression analysis.

To evaluate whether the variables satisfy the assumption of proportionality in the hazards model is that obtained by summing over individuals resulting in a process that varies over time and such residuals are called *Schoenfeld residuals*.

The Schoenfeld residuals at the $i - th$ observed event is given by:

$$s_i = \int_{t_{i-1}}^{t_i} \sum_{i'} \left[x'_i - \bar{x}(\hat{\beta}, s) \right] d\hat{M}_i(s)$$

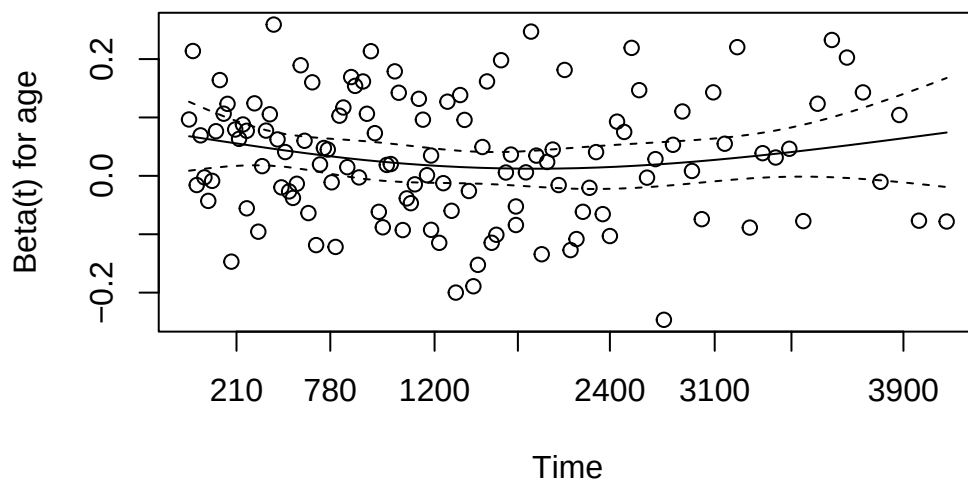
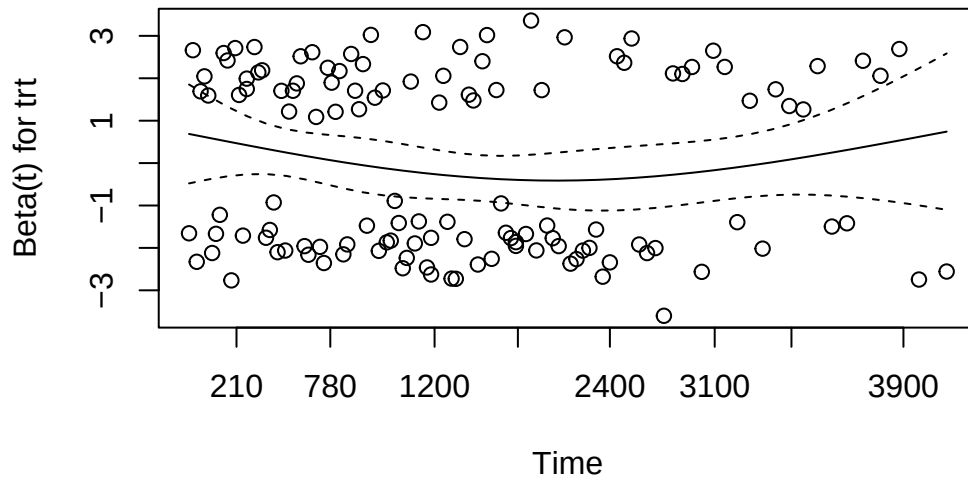
Example: Residual Analysis for the PBC Data

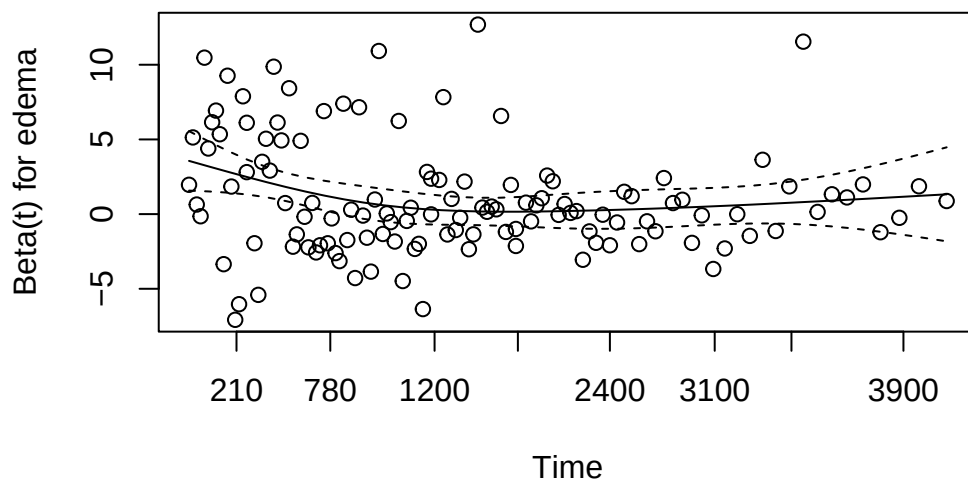
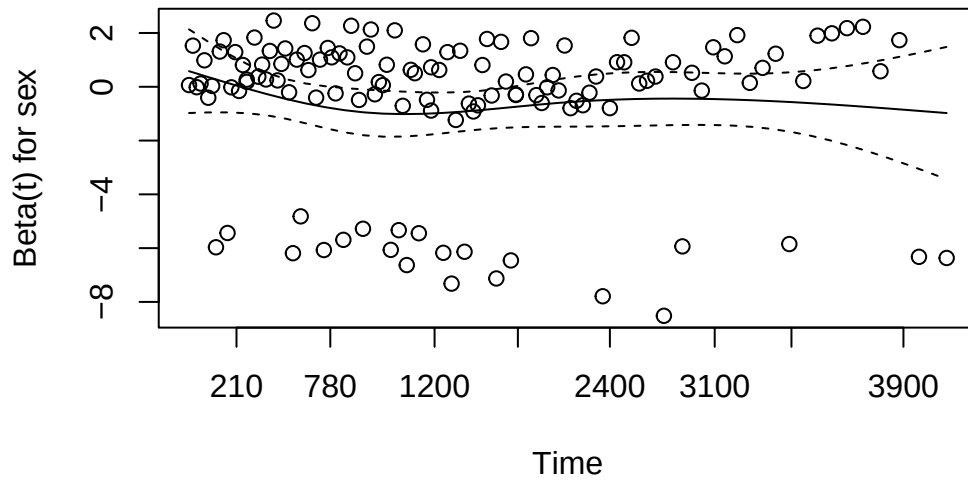
```
# Calculate residuals and run the PH test
ph_test <- cox.zph(pbc_ph)
print(ph_test)
```

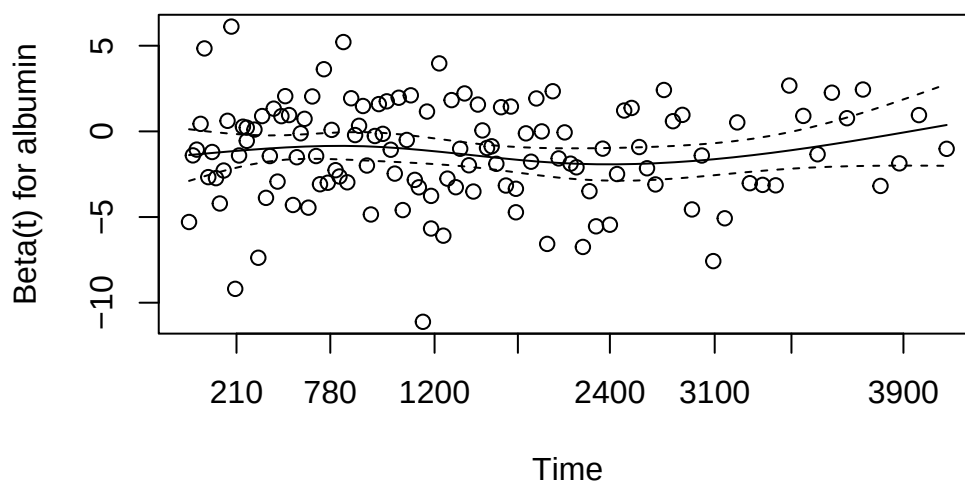
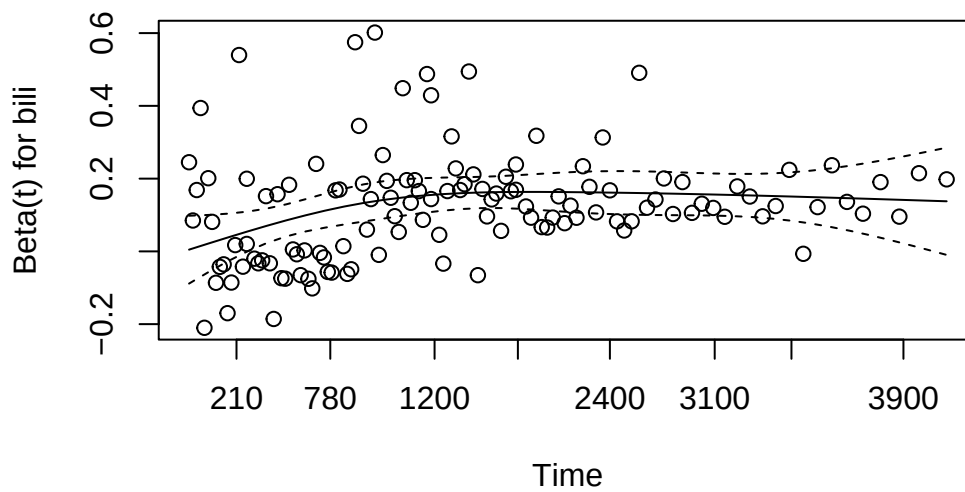
	chisq	df	p
trt	0.1310	1	0.71739
age	0.3034	1	0.58177
sex	0.0415	1	0.83865
edema	2.7823	1	0.09531
bili	12.0462	1	0.00052
albumin	0.2606	1	0.60974
GLOBAL	18.2115	6	0.00572

Plotting the residuals in R

```
# Plot all covariates
plot(ph_test)
```







Visual pattern: The solid trend line is flat and horizontal, hovering close to zero across the entire timeline. Interpretation: The hazard ratio for this variable is constant over time. The proportional hazards assumption is satisfied.

Parametric Regression Model

The Weibull distribution

Definition

Let T follow a Weibull distribution with shape parameter $p > 0$ and scale parameter $\lambda > 0$. One common parameterization is

$$F(t) = 1 - \exp \left[- \left(\frac{t}{\lambda} \right)^p \right], \quad t > 0.$$

Therefore,

$$S(t) = \exp \left[- \left(\frac{t}{\lambda} \right)^p \right].$$

The density is

$$f(t) = \frac{p}{\lambda} \left(\frac{t}{\lambda} \right)^{p-1} \exp \left[- \left(\frac{t}{\lambda} \right)^p \right],$$

and the hazard is

$$h(t) = \frac{p}{\lambda} \left(\frac{t}{\lambda} \right)^{p-1}.$$

Interpretation of the shape parameter

The Weibull hazard is monotonic:

$$\begin{aligned} p < 1 & : \text{hazard decreases with time,} \\ p = 1 & : \text{hazard is constant (exponential model),} \\ p > 1 & : \text{hazard increases with time.} \end{aligned}$$

This flexibility makes the Weibull distribution widely useful, although it cannot represent a hazard that first rises and later falls.

Weibull regression

Under an AFT formulation,

$$\log(T_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \sigma \varepsilon_i,$$

where ε_i has an extreme-value distribution.

For covariate x_j , the acceleration factor or time ratio is

$$AF_j = e^{\beta_j}.$$

Example interpretation Suppose

$$\hat{\beta}_{\text{treatment}} = 0.40.$$

Then

$$e^{0.40} = 1.49.$$

Holding other covariates fixed, the characteristic survival time for the treatment group is approximately 1.49 times that of the reference group—about 49% longer.

If instead $\hat{\beta} = -0.30$, then

$$e^{-0.30} = 0.741.$$

The characteristic survival time is about 74.1% of that of the comparison group, corresponding to about a 25.9% reduction.

Weibull regression in R The survival package provides `survreg()`, which fits parametric location-scale survival models. For Weibull models, R's `survreg()` uses an AFT parameterization.

```
library(survival)

data(lung)

# In the current lung data, status is coded 1=censored, 2=dead.
# Create a 0/1 event indicator for clarity.
lung$event <- as.integer(lung$status == 2)

fit.w <- survreg(
  Surv(time, event) ~ age + sex,
  data = lung,
  dist = "weibull"
)

summary(fit.w)
```

Call:

```
survreg(formula = Surv(time, event) ~ age + sex, data = lung,
  dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	6.27485	0.48137	13.04	< 2e-16
age	-0.01226	0.00696	-1.76	0.0781
sex	0.38209	0.12748	3.00	0.0027
Log(scale)	-0.28230	0.06188	-4.56	5.1e-06

Scale= 0.754

Weibull distribution

Loglik(model)= -1147.1 Loglik(intercept only)= -1153.9

Chisq= 13.59 on 2 degrees of freedom, p= 0.0011

Number of Newton-Raphson Iterations: 5

n= 228

```
coef(fit.w)
```

	age	sex
(Intercept)	6.27485306	-0.01225703 0.38208514

```
exp(coef(fit.w))
```

	age	sex
(Intercept)	531.0483429	0.9878178 1.4653368

```
# Time ratios for covariates only:
```

```
exp(coef(fit.w)[-1])
```

	age	sex
	0.9878178	1.4653368

If the coefficient for sex is positive, the group coded one unit higher has longer predicted survival times, conditional on the other covariates.

The log-logistic distribution

A useful parameterization of the log-logistic survival function is

$$S(t) = \frac{1}{1 + (t/\lambda)^p}, \quad t > 0, \quad \lambda, p > 0.$$

Therefore,

$$F(t) = \frac{(t/\lambda)^p}{1 + (t/\lambda)^p}.$$

The density is

$$f(t) = \frac{(p/\lambda)(t/\lambda)^{p-1}}{[1 + (t/\lambda)^p]^2},$$

and the hazard is

$$h(t) = \frac{(p/\lambda)(t/\lambda)^{p-1}}{1 + (t/\lambda)^p}.$$

Hazard shape

The log-logistic model is especially useful because its hazard can be non-monotonic.

- If $p \leq 1$, the hazard decreases with time.
- If $p > 1$, the hazard initially increases and later decreases.

Thus, it can describe situations where event risk rises after entry into the study, reaches a peak, and subsequently falls.

Log-logistic regression

The log-logistic AFT model can be written

$$\log(T_i) = \mathbf{x}_i^\top \beta + \sigma \varepsilon_i,$$

where ε_i follows a standard logistic distribution.

The coefficient interpretation is again conveniently expressed using

$$AF_j = e^{\beta_j}.$$

Thus, $\exp(\beta_j) > 1$ indicates longer survival times and $\exp(\beta_j) < 1$ indicates shorter survival times, holding other predictors constant.

The log-logistic model also has a proportional-odds representation for survival odds, which distinguishes it from the Weibull model.

Log-logistic regression in R

```
fit.ll <- survreg(
  Surv(time, event) ~ age + sex,
  data = lung,
  dist = "loglogistic"
```

```
)
```

```
summary(fit.ll)
```

Call:

```
survreg(formula = Surv(time, event) ~ age + sex, data = lung,  
        dist = "loglogistic")
```

	Value	Std. Error	z	p
(Intercept)	5.92232	0.53269	11.12	< 2e-16
age	-0.01401	0.00771	-1.82	0.06946
sex	0.47751	0.14036	3.40	0.00067
Log(scale)	-0.56991	0.06543	-8.71	< 2e-16

Scale= 0.566

Log logistic distribution

Loglik(model)= -1152.9 Loglik(intercept only)= -1160.9

Chisq= 16.07 on 2 degrees of freedom, p= 0.00032

Number of Newton-Raphson Iterations: 4

n= 228

```
# Time ratios  
exp(coef(fit.ll)[-1])
```

	age	sex
	0.9860925	1.6120541

Comparing Weibull and log-logistic models

For models fitted to the same observations, AIC is

$$AIC = -2\ell + 2k,$$

where ℓ is the maximized log-likelihood and k is the number of estimated parameters. Smaller AIC values indicate a better trade-off between fit and model complexity.

```
AIC(fit.w, fit.ll)
```

	df	AIC
fit.w	4	2302.109
fit.ll	4	2313.794

```
logLik(fit.w)
```

```
'log Lik.' -1147.054 (df=4)
```

```
logLik(fit.ll)
```

'log Lik.' -1152.897 (df=4)

AIC should not be the only consideration. The assumed hazard shape and scientific plausibility are also important.

Feature	Weibull	Log-logistic
Parametric	Yes	Yes
AFT formulation	Yes	Yes
Hazard shape	Monotonic	May be unimodal
Constant hazard possible	Yes ($p = 1$)	No general exponential case
Useful when risk rises then falls	No	Yes